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Lecture 07A
Before-Midterms-Session

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To obtain one point, you need to answer TRUTHFULLY.

0 - 3.75 points	4 - 7.75 points	8 - 11.75 points	12 - 15.75 points	16 - 20 points

Recall:

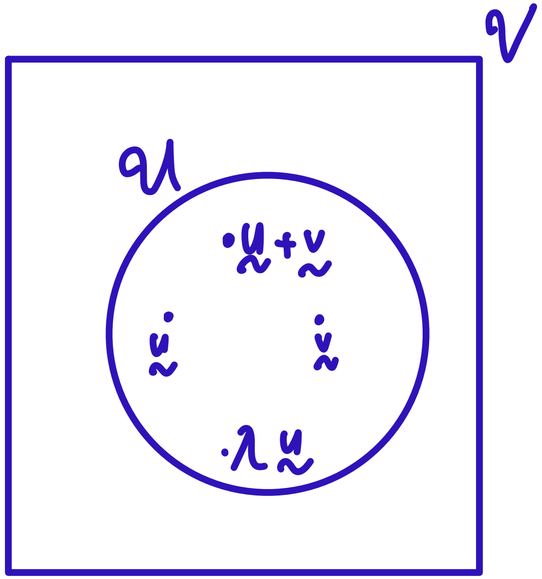
<u>Definition</u>. Let $\mathcal{V}$ be a vector space.

$\mathcal{U}$ is a *subspace* of $\mathcal{V}$

vector space

if $\mathcal{U}$ is a subset of $\mathcal{V}$ and $\mathcal{U}$ is a **BLANK**

Fill in the blank.



Question: Is $\mathcal{U}$ a subspace

This "$\mathcal{U}$" is big...

Recall:

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<u>Theorem</u>. (Conditions for Subspace) Let $\mathcal{V}$ be a vector space.

A subset $\mathcal{U}$ of $\mathcal{V}$ is a subspace if and only if $\mathcal{U}$ satisfies the following conditions.

- **(Zero Vector).** BLANK1 $\underset{\sim}{0}$ belongs to $\mathcal{U}$.
- **(Closure under** vector Addition BLANK2). For all $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{U}$, we have $\boldsymbol{u} \oplus \boldsymbol{v} \in \mathcal{U}$.
- **(Closure under** scalar multiplication BLANK3). For $\lambda \in \mathbb{F}$, $\boldsymbol{u} \in \mathcal{U}$, we have $\lambda \otimes \boldsymbol{u} \in \mathcal{U}$.

Fill in the blanks.

Midterms AY 2021/2022 Q1

For a matrix $\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n\times n}(\mathbb{R})$, and $i \in \{1, 2, \ldots, n\}$, let

$$r_i(\boldsymbol{A}) = a_{i1} + a_{i2} + \ldots + a_{in} \,.$$

In other words, $r_i(\boldsymbol{A})$ is the sum of entries in the ith row of $\boldsymbol{A}$.

(a) Consider the set $\mathcal{U} = \{\boldsymbol{A} \in \mathcal{M}_{n\times n}(\mathbb{R}) : \boldsymbol{r}_i(A) = 1 \text{ for all } i = 1, 2, \ldots, n\}$. Is $\mathcal{U}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

(b) Consider the set $\mathcal{V} = \{\boldsymbol{A} \in \mathcal{M}_{n\times n}(\mathbb{R}) : \boldsymbol{r}_i(A) \geq 0 \text{ for all } i = 1, 2, \ldots, n\}$. Is $\mathcal{V}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

(c) Consider the set $\mathcal{W} = \{\boldsymbol{A} \in \mathcal{M}_{n\times n}(\mathbb{R}) : \boldsymbol{r}_i(A) = 0 \text{ for some } i = 1, 2, \ldots, n\}$. Is $\mathcal{W}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

For a matrix $\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n \times n}(\mathbb{R})$, and $i \in \{1, 2, \ldots, n\}$, let

$$r_i(\boldsymbol{A}) = a_{i1} + a_{i2} + \ldots + a_{in} \, .$$

In other words, $r_i(\boldsymbol{A})$ is the sum of entries in the ith row of $\boldsymbol{A}$.

What is the zero vector in $\mathcal{M}_{n \times n}(\mathbb{R})$?

→ zero matrix

$$\begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

For a matrix $\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n\times n}(\mathbb{R})$, and $i \in \{1, 2, \ldots, n\}$, let

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In other words, $r_i(\boldsymbol{A})$ is the sum of entries in the ith row of $\boldsymbol{A}$.

(a) Consider the set $\mathcal{U} = \{\boldsymbol{A} \in \mathcal{M}_{n\times n}(\mathbb{R}) : \boldsymbol{r}_i(A) = 1 \text{ for all } i = 1, 2, \ldots, n\}$. Is $\mathcal{U}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

Is the zero vector in $\mathcal{U}$?

No.

$r_i(\hat{0}) = 0 \neq 1$

$\Rightarrow \hat{0} \notin \mathcal{U}$

For a matrix $\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n\times n}(\mathbb{R})$, and $i \in \{1, 2, \ldots, n\}$, let

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(a) Consider the set $\mathcal{U} = \{\boldsymbol{A} \in \mathcal{M}_{n\times n}(\mathbb{R}) : \boldsymbol{r}_i(A) = 1 \text{ for all } i = 1, 2, \ldots, n\}$. Is $\mathcal{U}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

Is $\mathcal{U}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

No.

In other words, $r_i(A)$ is the sum of entries in the ith row of $\boldsymbol{A}$.

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(b) Consider the set $\mathcal{V} = \{A \in \mathcal{M}_{n\times n}(\mathbb{R}) : r_i(A) \geq 0 \text{ for all } i = 1, 2, \ldots, n\}$. Is $\mathcal{V}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

Is Albert's proof that "$\mathcal{V}$ is closed under scalar multiplication" correct? No. Shouldn't give specific numbers

Albert's Solution: Let $\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \in \mathcal{V}$ and choose $\lambda = 1048576$.

Then

$$\lambda\boldsymbol{A} = 1048576 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1048576 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Hence, $\lambda\boldsymbol{A}$ belongs to $\mathcal{V}$ too.

For a matrix $\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n\times n}(\mathbb{R})$, and $i \in \{1, 2, \ldots, n\}$, let

$$r_i(\boldsymbol{A}) = a_{i1} + a_{i2} + \ldots + a_{in} .$$

In other words, $r_i(\boldsymbol{A})$ is the sum of entries in the ith row of $\boldsymbol{A}$.

(b) Consider the set $\mathcal{V} = \{\boldsymbol{A} \in \mathcal{M}_{n\times n}(\mathbb{R}) : r_i(\boldsymbol{A}) \geq 0$ for all $i = 1, 2, \ldots, n\}$. Is $\mathcal{V}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

Is $\mathcal{V}$ closed under scalar multiplication?

No. $\lambda = -1$ will caused it not be closed under scalar multiplication.

Choose $\hat{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $\lambda = -1$

then $\lambda\hat{A} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \notin \mathcal{V}$

$\therefore \mathcal{V}$ is not closed under $\times$

For a matrix $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n\times n}(\mathbb{R})$, and $i \in \{1, 2, \ldots, n\}$, let

$$r_i(A) = a_{i1} + a_{i2} + \ldots + a_{in} .$$

In other words, $r_i(A)$ is the sum of entries in the ith row of A.

(b) Consider the set $\mathcal{V} = \{A \in \mathcal{M}_{n\times n}(\mathbb{R}) : r_i(A) \geq 0 \text{ for all } i = 1, 2, \ldots, n\}$. Is $\mathcal{V}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

Is $\mathcal{V}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

No. $\mathcal{V}$ is not closed under scalar multiplication. $\therefore$ $\mathcal{V}$ is NOT a subspace.

For a matrix $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n\times n}(\mathbb{R})$, and $i \in \{1, 2, \ldots, n\}$, let

$$r_i(A) = a_{i1} + a_{i2} + \ldots + a_{in}\,.$$

In other words, $r_i(A)$ is the sum of entries in the ith row of A.

(c) Consider the set $\mathcal{W} = \{A \in \mathcal{M}_{n\times n}(\mathbb{R}) : r_i(A) = 0 \text{ for some } i = 1, 2, \ldots, n\}$. Is $\mathcal{W}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

Which of the following matrices belong to $\mathcal{W}$?

1. $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ $r_1 = 0$, $r_2 = r_3 = 3$ $\Rightarrow$ belongs to $\mathcal{W}$

2. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $r_1 = 3$, $r_2 = r_3 = 0$ $\Rightarrow$ belongs to $\mathcal{W}$

3. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ $r_1 = r_2 = r_3 = 3$ $\Rightarrow$ not in $\mathcal{W}$

For a matrix $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n\times n}(\mathbb{R})$, and $i \in \{1, 2, \ldots, n\}$, let

$$r_i(A) = a_{i1} + a_{i2} + \ldots + a_{in}\,.$$

In other words, $r_i(A)$ is the sum of entries in the ith row of A.

(c) Consider the set $\mathcal{W} = \{A \in \mathcal{M}_{n\times n}(\mathbb{R}) : r_i(A) = 0 \text{ for some } i = 1, 2, \ldots, n\}$.

Is Albert's proof that "$\mathcal{W}$ is closed under vector addition" correct? No.

<u>Albert's Solution</u>: Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be matrices in $\mathcal{W}$.

Then there is some i such that

$$r_i(\boldsymbol{A}) = r_i(\boldsymbol{B}) = 0.$$

i may not be the same!

In other words, the i-th rows of $\boldsymbol{A}$ and $\boldsymbol{B}$ sum to zero.

Then $r_i(\boldsymbol{A}+\boldsymbol{B}) = r_i(\boldsymbol{A}) + r_i(\boldsymbol{B}) = 0.$

Hence, $\boldsymbol{A}+\boldsymbol{B}$ belongs to $\mathcal{W}$ too.

For a matrix $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n\times n}(\mathbb{R})$, and $i \in \{1, 2, \ldots, n\}$, let

$$r_i(A) = a_{i1} + a_{i2} + \ldots + a_{in}\,.$$

In other words, $r_i(A)$ is the sum of entries in the ith row of A.

(c) Consider the set $\mathcal{W} = \{A \in \mathcal{M}_{n\times n}(\mathbb{R}) : r_i(A) = 0 \text{ for some } i = 1, 2, \ldots, n\}$. Is $\mathcal{W}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

Is $\mathcal{W}$ closed under vector addition?

No.

Counter example:

Choose $\hat{A} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, $\hat{B} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $\hat{A}$ and $\hat{B} \in \mathcal{W}$

$\hat{A} + \hat{B} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \notin \mathcal{W}$

For a matrix $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \in \mathcal{M}_{n\times n}(\mathbb{R})$, and $i \in \{1, 2, \ldots, n\}$, let

$$r_i(A) = a_{i1} + a_{i2} + \ldots + a_{in} \,.$$

In other words, $r_i(A)$ is the sum of entries in the ith row of A.

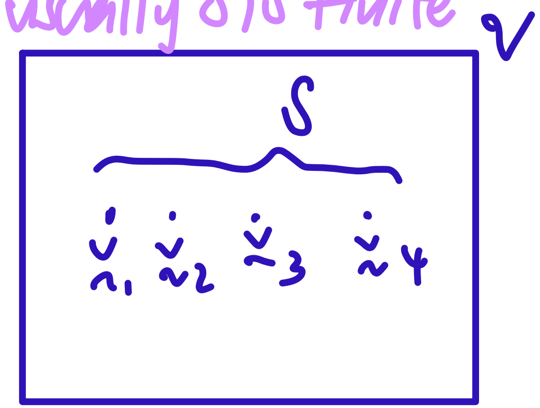
(c) Consider the set $\mathcal{W} = \{A \in \mathcal{M}_{n\times n}(\mathbb{R}) : r_i(A) = 0 \text{ for some } i = 1, 2, \ldots, n\}$. Is $\mathcal{W}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

Is $\mathcal{W}$ a subspace of $\mathcal{M}_{n\times n}(\mathbb{R})$?

No. Because $\mathcal{W}$ is NOT closed under $\oplus$.

Recall:

usually S is finite



Definition. Let $\mathcal{V}$ be a vector space.

Let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$ be a subset of $\mathcal{V}$.

(a) If all vectors in $\mathcal{V}$ can be written as a linear combination of vectors in S, then we say BLANK1 S spans $\mathcal{V}$ $\Rightarrow$ span$(S) = \mathcal{V}$.

(b) If $\lambda_1 \boldsymbol{v}_1 + \cdots + \lambda_n \boldsymbol{v}_n = \boldsymbol{0}$ implies $\lambda_1 = \cdots = \lambda_n = 0$, then we say that BLANK2 linearly independent

Question: Does S span $\mathcal{V}$?

Is S lin. ind?

Fill in the blanks.

· S is a basis $\Longleftrightarrow$ S is L.I. & span$(S) = \mathcal{V}$

Question 3. Let $\mathcal{V}$ be a real vector space. Suppose $\boldsymbol{u}$ and $\boldsymbol{v}$ are linearly independent vectors in $\mathcal{V}$. Find condition(s) on real numbers a, b, c and d such that $\{a\boldsymbol{u}+b\boldsymbol{v}, c\boldsymbol{u}+d\boldsymbol{v}\}$ is linearly independent. Is $\{\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{u}-\boldsymbol{v}\}$ linearly independent?

Tut03 Q3 (simplified). Suppose that $\{\boldsymbol{u}, \boldsymbol{v}\}$ is linearly independent. Show that $\{\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{u}-\boldsymbol{v}\}$ is linearly independent.

Tut03 Q3 (simplified). Suppose that $\{\boldsymbol{u}, \boldsymbol{v}\}$ is linearly independent. Show that $\{\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{u}-\boldsymbol{v}\}$ is linearly independent.

What is the $(\star)$ that corresponds to showing $\{\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{u}-\boldsymbol{v}\}$ is linearly independent?

1. $\lambda \boldsymbol{u} + \mu \boldsymbol{v} = \boldsymbol{0}$

2. ✓ $\lambda\boxed{(\boldsymbol{u}+\boldsymbol{v})} + \mu(\boldsymbol{u}-\boldsymbol{v}) = \boldsymbol{0}$

WTS: $\{\square, \triangle\}$ is L.I.

$\lambda \square + \mu \triangle$ — $(\star)$

$\Rightarrow \lambda = \mu = 0$ $\Rightarrow$ WTS this

Tut03 Q3 (simplified). Suppose that $\{\boldsymbol{u}, \boldsymbol{v}\}$ is linearly independent. Show that $\{\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{u}-\boldsymbol{v}\}$ is linearly independent.

$$\lambda(\underset{\sim}{u} + \underset{\sim}{v}) + \mu(\underset{\sim}{u} - \underset{\sim}{v}) = 0 \quad — (*)$$

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Now,

$$\lambda(\boldsymbol{u}+\boldsymbol{v}) + \mu(\boldsymbol{u}-\boldsymbol{v}) = (\lambda+\mu)\boldsymbol{u} + (\lambda-\mu)\,\boldsymbol{v}$$

Why is this true?

"Regrouping terms"

Tut03 Q3 (simplified). Suppose that $\{\boldsymbol{u}, \boldsymbol{v}\}$ is linearly independent. Show that $\{\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{u}-\boldsymbol{v}\}$ is linearly independent.

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$$(\lambda+\mu)\boldsymbol{u}+(\lambda-\mu)\boldsymbol{v}=\mathbf{0}$$

Implies that $\lambda+\mu = \lambda-\mu = 0$

Why is this true?

Because $\{\underset{\sim}{u}, \underset{\sim}{v}\}$ is L. I.

(This is true by assumption.)

Tut03 Q3 (simplified). Suppose that $\{\boldsymbol{u}, \boldsymbol{v}\}$ is linearly independent. Show that $\{\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{u}-\boldsymbol{v}\}$ is linearly independent.

$$\lambda + \mu = \lambda - \mu = 0$$

Implies that $\lambda = \mu = 0$.

$$\begin{cases} \lambda + \mu = 0 \\ \lambda - \mu = 0 \end{cases} \Rightarrow \lambda = \mu = 0$$

Why is this true?

We are solving two equations in the two unknowns λ and μ.

Tut03 Q3 (simplified). Suppose that $\{\boldsymbol{u}, \boldsymbol{v}\}$ is linearly independent. Show that $\{\boldsymbol{u}+\boldsymbol{v}, \boldsymbol{u}-\boldsymbol{v}\}$ is linearly independent.

$$\lambda(\underset{\sim}{u}+\underset{\sim}{v}) + \mu(\underset{\sim}{u}-\underset{\sim}{v}) = \underset{\sim}{0} \qquad (*)$$

$$(\lambda+\mu)\underset{\sim}{u} + (\lambda-\mu)\underset{\sim}{v} = \underset{\sim}{0}$$

$\Rightarrow \lambda+\mu = \lambda-\mu = 0$ Because $\{\underset{\sim}{u}, \underset{\sim}{v}\}$ is L.I.

$\Rightarrow \lambda = \mu = 0$

$\therefore \{\underset{\sim}{u}+\underset{\sim}{v}, \underset{\sim}{u}-\underset{\sim}{v}\}$ is L.I.

Recall:

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Definition. Let $\mathcal{V}$ be a vector space.

Let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_n\}$ be a subset of $\mathcal{V}$.

S is a basis for V

If $\text{span}(S) = \mathcal{V}$ and S is linearly independent, then **BLANK1**

number of vectors in S

The dimension of $\mathcal{V}$ is given by the **BLANK2** in a basis for $\mathcal{V}$.

Revise the conditions to check when S is a basis

(check number is correct).

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Follow the link to the seating arrangement for **Midterms**.

The same link is in Telegram.

Can you locate your name?